

MTH212M PRACTICE PROBLEMS WEEK 3

Question 1. Fix $p \in (0, 1)$ and let $\{X_n\}_{n \geq 1}$ be an i.i.d. sequence of *Bernoulli*(p) RVs. Define $Y_n := \min\{10, X_1 + X_2 + \cdots + X_n\}$, $n = 1, 2, \dots$. Identify the recurrent/transient states for the Markov Chain $\{Y_n\}_{n \geq 1}$.

Question 2. A urn contains b black balls and w white balls. Draw a ball at random, and replace it by a ball of the other colour. Let X_n denote the number of black balls after the n -th draw. Identify the recurrent/transient states for the Markov Chain $\{X_n\}_{n \geq 1}$.

Question 3. Consider a Markov chain $\{X_n\}_{n=0}^\infty$ with state space S and the transition probability matrix P . Find $\lim_{n \rightarrow \infty} p_{ii}^n, \forall i \in S$.

$$(i) \ S = \{1, 2, 3, 4\} \text{ and } P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

$$(ii) \ S = \{1, 2, 3, 4, 5\} \text{ and } P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$